Frozen FRAMES: initialization and sampling robustness

No new parameter search • Noiseless rank-one n=2 • Previously exposed channels 2–7

12/12 audited trajectories pass; 12 completed, 0 numerical failures; 0 planned trajectories unaudited.

Channel	Init.	Stream	RMSE	TP norm.	Relative error	Infidelity (1−F)	Gate
2	1	10	7.76e-07	6.86e-06	4.68e-06	1.92e-11	PASS
2	1	11	7.34e-07	6.76e-06	4.43e-06	1.16e-11	PASS
3	1	10	7.13e-07	6.61e-06	4.31e-06	1.37e-11	PASS
3	1	11	6.82e-07	6.29e-06	4.12e-06	1.87e-11	PASS
4	1	10	7.45e-07	6.67e-06	4.56e-06	2.32e-11	PASS
4	1	11	7.66e-07	6.60e-06	4.69e-06	3.91e-11	PASS
5	1	10	6.53e-07	5.96e-06	3.92e-06	1.20e-11	PASS
5	1	11	6.28e-07	5.76e-06	3.77e-06	8.91e-12	PASS
6	1	10	6.47e-07	5.87e-06	3.89e-06	2.21e-11	PASS
6	1	11	6.03e-07	5.47e-06	3.63e-06	1.25e-11	PASS
7	1	10	6.96e-07	6.50e-06	4.18e-06	1.11e-11	PASS
7	1	11	6.92e-07	6.49e-06	4.15e-06	1.20e-11	PASS

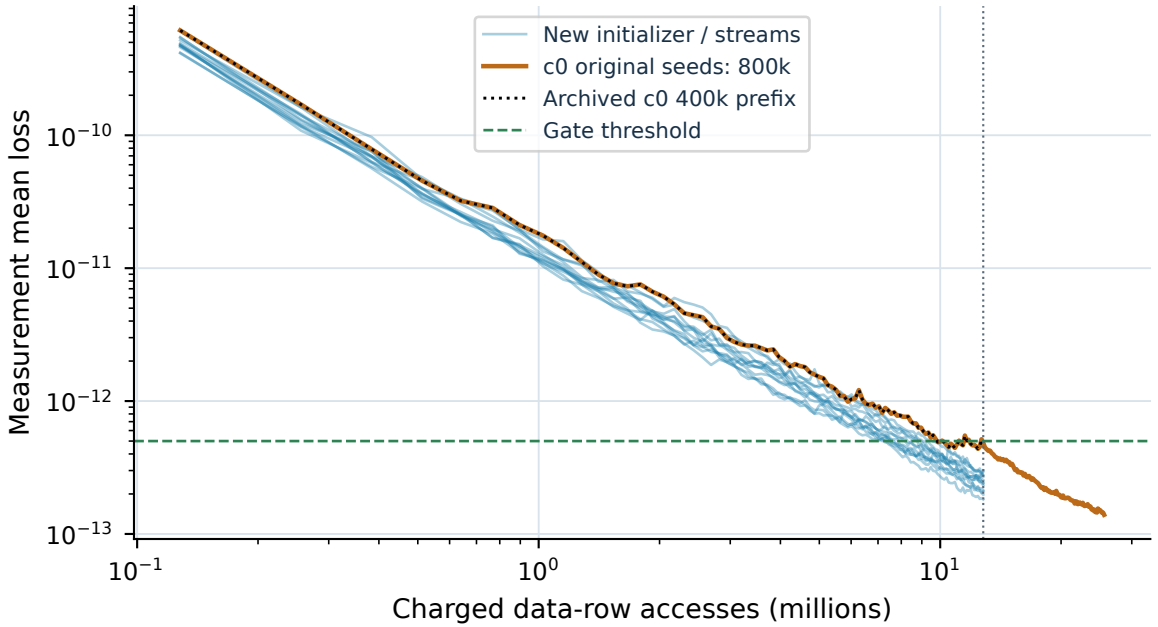
Six channels × one new initializer × two new sampling streams = twelve trajectories, not twelve independent channels.
Initialization and sampling both change: this tests joint robustness, not an isolated causal initialization effect.

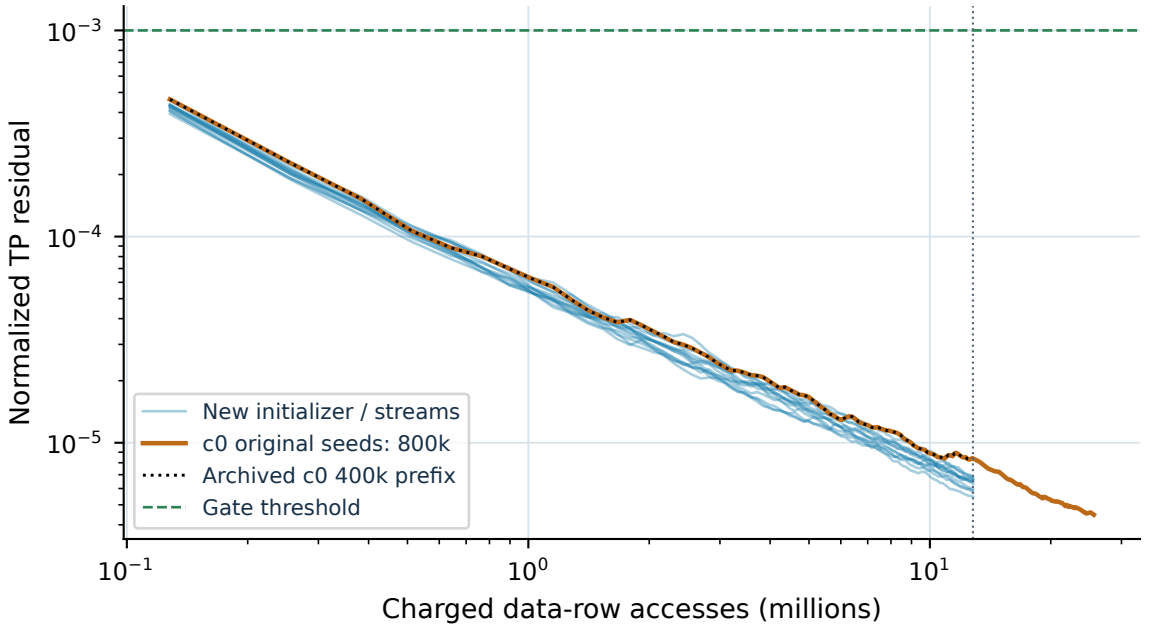
Frozen controls: alpha0=5e-5; alpha(t)=alpha0(t+1)^0.25; beta(t)=1/[2alpha(t)]; gamma=min(1,4/(t+4)); rho=min(1,4/(t+8)^(2/3)); batch32; tau=sqrt(d). Every robustness trajectory restarts for 400,000 updates; all work is counted.

Unchanged joint gate: fidelity >=0.99, full-design RMSE <=1e-6 and TPnorm <=1e-3. Outcomes do not trigger retuning. Final held-out channels 1000–1019 remain excluded.

Measurement loss, TP error and the longer channel-0 tail

Measurement loss excludes the TP penalty • Charged rows exclude TP, LMO, metrics and compilation work





Tail endpoint: PASS; RMSE 5.28e-07, TPnorm 4.47e-06. Beyond 400k: 100/100 recorded checkpoints pass. Last failing checkpoint after 400k: none.

Independent 400k prefix audit: MATCH; maximum RMSE difference 0.00e+00. Stored physical metrics, schedules and sampled batches are compared; intermediate factors are not archived. The longer run starts afresh, so its duplicated 400k prefix is charged. Checks every 4,000 updates support only observed checkpoint stability, not continuous-time or asymptotic convergence.

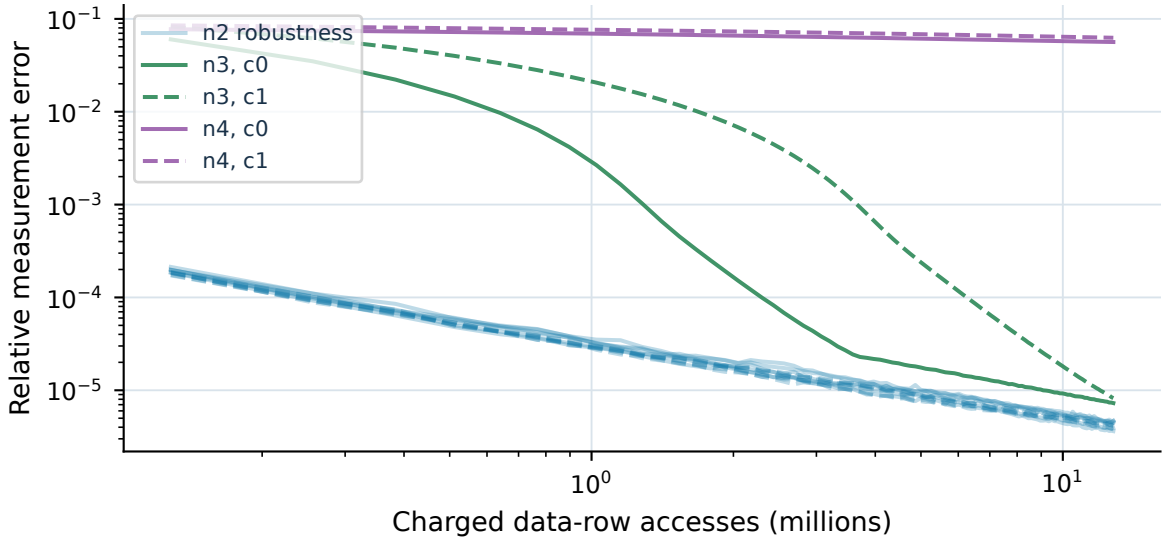
Channel 0 first passed at 308k but later crossed back above the gate; its 400k endpoint had a narrow RMSE margin. This follow-up checks that same seed trajectory beyond the old endpoint; it is not another independent channel. The zero-row initial point is omitted on logarithmic axes.

Frozen-parameter dimension transfer: exploratory only

n=3/4 use unchanged structured NumPy FRAMES; n=2 uses the pinned JAX CPU adapter

Stage	n	Chan.	Updates	Row epochs	RMSE	Rel. error	TP norm.	1−F	Gate
pilot	3	0	2,000	62.5	7.08e-03	8.49e-02	2.65e-02	5.13e-01	FAIL
pilot	4	0	2,000	7.8125	3.44e-03	8.12e-02	1.11e-02	9.89e-01	FAIL
transfer	3	0	400,000	12500	6.06e-07	7.28e-06	7.65e-06	1.10e-10	PASS
transfer	3	1	400,000	12500	6.82e-07	8.24e-06	8.49e-06	1.13e-10	PASS
transfer	4	0	400,000	1562.5	2.39e-03	5.64e-02	1.33e-03	8.13e-01	FAIL
transfer	4	1	400,000	1562.5	2.66e-03	6.26e-02	1.65e-03	8.80e-01	FAIL

All follow-up stages: 19/19 completed; 0 numerical failures; 0 unaudited. Pilot gate outcomes are short-budget diagnostics, not recovery tests.



n3: 400,000-update budget; n4: 400,000-update budget. Budget decisions use pilot runtime/memory only, never numerical success.

Relative error = $||A(UU^H)-y||_2 / ||y||_2$. The same absolute RMSE threshold does not mean the same relative accuracy across dimensions. Alpha0 is preserved per mean-loss row; summed lambda, POVM constants, tau and row epochs change with dimension.

Both n4 cases failed data fit, fidelity and TP, not just TP.
Next: separately declare n4 penalty/sampling tuning. Quiroga remains unresolved; no head-to-head or speed claim.

Noiseless rank-one tests only; no noisy or higher-rank conclusions. Regression suite: 93 tests and 7 subtests passed. Source 4ce1108a0692 and all input hashes are in the companion evidence JSON.